

# Four dimensions are minimal

**Verdict.** Over the complex numbers, the least ambient dimension is **4** for an evolution algebra  $E$  whose stable derived term  $D$  satisfies  $N(D) = \{0\}$  but admits no natural basis. The earlier four-dimensional witness supplies the upper bound; the elementary argument below supplies the lower bound.

This is a source-informed, AI-assisted follow-up commissioned by Mark S. Weinstein, in the same assistant workflow as the earlier audit. It is a reconstructed proof, with supporting exact calculations, not a claim of historical novelty or independent human expert review.

## 1. Definitions and stable-term closure

A natural basis has zero products between distinct basis vectors. Multiplication is bilinear and commutative; associativity is not assumed. Write  $V^2 = \text{span}\{uv : u,v \in V\}$ ,  $E^{(1)} = E$  and  $E^{(k+1)} = (E^{(k)})^2$ . Also  $N(D) = \{d \in D : d^2 = 0\}$ ; this is a set, not an assumed vector subspace.

If  $W$  is a subalgebra,  $W^2 \subseteq W$  and products of elements of  $W^2$  lie in  $W^2$ . Thus each derived term is a subalgebra and the series descends. Finite dimension gives stabilization, so  $D$  is a commutative subalgebra and  $D^2 = D$ . We make no assertion that  $D$  has a natural basis inherited from  $E$ .

## 2. The low-dimensional lemma

**Lemma.** Every commutative complex algebra  $A$  of dimension at most two with  $N(A) = \{0\}$  admits a natural basis. In dimensions zero and one, respectively use the empty basis and any single basis vector.

**Proof in dimension two.** Choose a basis  $a,b$  and let  $L_u(v) = uv$ . If  $L_a$  is singular, take  $u = a$ . Otherwise  $\det(L_b + tL_a)$  is a quadratic in  $t$  with nonzero leading coefficient  $\det(L_a)$ . Since  $C$  is algebraically closed, it has a root  $t_0$ . Take  $u = b + t_0a \neq 0$ ; then  $L_u$  is singular.

Choose  $0 \neq v \in \ker L_u$ . Since  $u^2 \neq 0$ ,  $u$  is not in this kernel, and  $u,v$  are independent. Their mixed product  $uv$  is zero. Commutativity therefore makes  $\{u,v\}$  a natural basis. This proves the lemma without assuming  $A^2 = A$  or importing a classification theorem.

## 3. Every smaller ambient case is excluded

| dim $D$ , when $\dim E \leq 3$ | Reason the target cannot hold                                          |
|--------------------------------|------------------------------------------------------------------------|
| 0                              | The empty basis makes the zero algebra an evolution algebra.           |
| 1                              | Every one-dimensional algebra has a natural basis.                     |
| 2                              | $N(D) = \{0\}$ gives a natural basis by the lemma.                     |
| 3                              | Then $\dim E = 3$ and $D = E$ , which is already an evolution algebra. |

Equivalently, the target forces  $\dim D \geq 3$  and  $D \neq E$ , so  $\dim E \geq \dim D + 1 \geq 4$ . The convention including the zero algebra via its empty natural basis is explicit here.

# The witness, reconstructed

The structure matrix below is the construction in the earlier audit, p. 8, and published checker C05-C06. It is carried forward and rechecked here; the minimality proof on p. 1 is the additional conclusion of this follow-up.

Let  $E$  have natural basis  $e_1, e_2, e_3, e_4$  with  $e_i e_j = 0$  for  $i \neq j$ . Row  $i$  of  $W$  gives  $e_i^2$ , so all multiplication conventions match the published record.

$$W = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

In particular,  $e_1^2 = e_1 - e_3$ ;  $e_2^2 = e_1 + e_2 + e_4$ ;  $e_3^2 = e_2 - e_3 - e_4$ ; and  $e_4^2 = e_3 + e_4$ .

## 4. Its stable term has dimension three

Put  $f_1 = e_1 + e_4$ ,  $f_2 = e_2$ ,  $f_3 = e_3 + e_4$ . All four rows lie in the hyperplane with fourth coordinate equal to the sum of the first and third. They span it:  $f_3 = e_4^2$ ,  $f_1 = e_1^2 + e_4^2$  and  $f_2 = e_2^2 - f_1$ . Consequently

$$E^2 = D = \{(x, y, z, x+z) : x, y, z \in C\}.$$

Identify  $D$  with  $C^3$  through  $(x, y, z) \mapsto xf_1 + yf_2 + zf_3$ . Direct bilinear multiplication from  $W$  gives

$$(x, y, z)(u, v, w) = (xu + yv, yv + zw, xw + zu).$$

The products  $f_1^2 = f_1$ ,  $f_2^2 = f_1 + f_2$  and  $f_1 f_3 = f_3$  span  $D$ . Hence  $D^2 = D$ , and the derived dimensions are  $4 \rightarrow 3 \rightarrow 3$ . Here  $E^2$  is already the stable derived term.

## 5. Every square-zero solution is covered

The induced square map is

$$Q(x, y, z) = (x^2 + y^2, y^2 + z^2, 2xz).$$

If  $Q(x, y, z) = 0$ , then  $xz = 0$  because  $2 \neq 0$  in  $C$ . If  $x = 0$ , the first equation forces  $y = 0$  and the second forces  $z = 0$ . If  $z = 0$ , the second equation forces  $y = 0$  and the first forces  $x = 0$ . Those two cases cover every complex solution. Thus  $\mathbf{N}(D) = \{0\}$ .

No bounded enumeration or approximation of complex solutions is used. Checks M01-M03 confirm the exact embedding, spanning products and polynomial coefficients; the exhaustive implication above is a mathematical argument.

# Its stable term has no natural basis

## 6. A necessary condition gives a contradiction

The three coordinate bilinear forms of the product on p. 2 have matrices

$B_1$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$B_2$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$B_3$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Let  $B_0 = B_1 + B_2 = \text{diag}(1, 2, 1)$ , which is invertible. Set  $A = B_0^{-1}B_1 = \text{diag}(1, 1/2, 0)$  and  $C = B_0^{-1}B_3$ . Their commutator is

$$AC - CA = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

Suppose  $D$  had a natural basis whose coordinate columns form an invertible matrix  $P$ . Every mixed basis product would vanish in every output coordinate, so every  $P^T B_j P$  ( $j = 1, 2, 3$ ) would be diagonal. Their sum  $P^T B_0 P$  would also be invertible diagonal.

For each  $j$ , the following identity then makes the left-hand side diagonal:

$$P^{-1}(B_0^{-1}B_j)P = (P^T B_0 P)^{-1}(P^T B_j P).$$

In particular,  $A$  and  $C$  would become diagonal under the same similarity and would commute. The displayed nonzero commutator contradicts that consequence. Therefore  **$D$  admits no natural basis**. This uses only a necessary condition, not an imported simultaneous-diagonalization classification. M04 verifies the finite matrix data exactly.

Combining pp. 1-3 establishes both the lower and upper bounds. **The minimum is exactly four.** The same witness shows that dimension three is the least intrinsic dimension of a commutative complex square-zero-free algebra without a natural basis.

## 7. A useful consequence for the sufficient conditions

For any complex evolution algebra of ambient dimension at most three, the conditions  **$N(D) = \{0\}$**  and  **$D$  is an evolution algebra** coincide. The forward direction is the dimension argument on p. 1.

For the converse in any finite dimension, if  $D$  has a natural basis  $g_1, \dots, g_m$ , its basis squares span  $D$  because  $D^2 = D$ . Those  $m$  squares are therefore independent. For  $d = \sum c_i g_i$ , the equation  $d^2 = \sum c_i^2 g_i^2 = 0$  forces every  $c_i = 0$ . The zero-dimensional case is vacuous. This is Barei's Remark 3.6, rederived here. The first strict separation is attained by the four-dimensional witness.

# Degenerate cases and exact checks

## 8. What the short lemma covers

**Singular multiplication operators.** The singular branch in the lemma is intentional. Under  $N(A) = \{0\}$ , a nonzero  $u$  cannot have  $L_u = 0$  because that would give  $u^2 = 0$ . A singular  $L_u$  in dimension two therefore has rank one, and its kernel supplies the second natural-basis vector. No assumption that some  $L_u$  is invertible is required.

**Degenerate product image.** If  $\dim A^2 = 0$  in a nonzero algebra, every square is zero. If  $\dim A^2 = 1$  in dimension two, squares have the form  $q(x,y)c$  for a fixed nonzero  $c$  and a homogeneous binary quadratic  $q$ . If  $q = 0$ , every vector is square-zero. Otherwise  $q$  has a nonzero complex zero: if its  $x^2$  coefficient is zero,  $(1,0)$  is a zero; if that coefficient is nonzero,  $q(t,1)$  has a complex root. Thus both product-image cases violate  $N(A) = \{0\}$ .

**Field boundary.** Over  $R$ , the algebra with  $a^2 = a$ ,  $ab = b$ ,  $b^2 = -a$  has no nonzero square-zero element, since a square has coordinates  $(x^2-y^2, 2xy)$ . But  $\det L_{xa+yb} = x^2+y^2 > 0$  for every nonzero real vector; no two independent real vectors can have zero product. Over  $C$ , the determinant roots  $\pm i$  instead give a natural basis. This is a deliberate change of field, not a counterexample to the lemma.

**Removing the square-zero hypothesis.** The complex dual numbers  $a^2 = a$ ,  $ab = b$ ,  $b^2 = 0$  have no natural basis: if  $uv = 0$  and either vector has nonzero  $a$ -coordinate, that vector's multiplication operator is invertible and the other vector is zero; otherwise both vectors lie on  $Cb$ . Thus no independent zero-product pair exists.

## 9. Actual supporting computations

| Check | Exact data or control verified                                        |
|-------|-----------------------------------------------------------------------|
| M01   | Nine witness embedding and basis-product identities.                  |
| M02   | Derived dimensions [4,3,3] and spanning products.                     |
| M03   | Square-map coefficients $(x^2+y^2, y^2+z^2, 2xz)$ .                   |
| M04   | $B_0$ inverse and nonzero commutator.                                 |
| M05   | Formal determinant identity with six independent structure constants. |
| M06   | Initially singular operator yields independent zero-product vectors.  |
| M07   | Nonreal roots and basis verified exactly over rational pairs $Q(i)$ . |
| M08   | Zero-image and rank-one product-image controls.                       |
| M09   | Dual-number square-zero and determinant data.                         |

Python 3.12.14, exit code 0: **9/9 groups PASS**. The unchanged published checker also ran with **11/11 PASS**; its eleven check records match the published results. Actual executions: 3 October 2026, 20:04:29 UTC (4:04:29 p.m. EDT). The programs check exact finite data and a formal polynomial identity, not the entire universal proof.

# Sources, addendum, and review package

## 10. Exact source locations and dependencies

Current main inspected at commit **a52a1caf3c7e2f560eb9c3ee01f676153d299f83**. The local detailed report, published note, checking code and recorded results match the pinned tree's Git blob identities and sizes.

Earlier audit: [detailed report](#) p. 8, construction and obstruction; p. 4, stable-term closure. [Published checker lines 152-184](#), C05-C06; its actual results are in `evidence/results_rechecked.json`. These establish the carried-forward witness, now rechecked.

[Barei's official paper](#): natural-basis definition, p. 1; §2, derived series and  $N(V)$ , p. 2; Corollary 3.4, pp. 3-4; Remark 3.6, p. 4. [Hu-Wen](#): natural basis and row-matrix convention, p. 1; Proposition 2.8, p. 6. Both complete primary manuscripts were retrieved for this follow-up.

The low-dimensional lemma and lower bound are proved here, not attributed to an imported classification or idempotent-existence theorem. The only root-existence input is the stated algebraic closure of  $C$ ; the remaining steps are elementary linear algebra and bilinearity. No essential imported mathematical dependency is unresolved. Historical priority for this lemma or minimality statement was not established.

## 11. Proposed addendum to the existing audit

**Minimal ambient dimension of the strictness witness.** The four-dimensional construction in this audit has the least possible ambient dimension over  $C$  among evolution algebras whose stable term  $D$  has  $N(D) = \{0\}$  but no natural basis.

Indeed, any commutative complex algebra  $A$  of dimension two with  $N(A) = \{0\}$  admits a natural basis. For a basis  $a, b$ , write  $L_u(v) = uv$ . If  $L_a$  is singular, take  $u = a$ ; otherwise the quadratic  $\det(L_b + tL_a)$  has a complex root  $t_0$ , giving nonzero  $u = b + t_0a$  with  $L_u$  singular. Choose  $0 \neq v$  in its kernel. Since  $u^2 \neq 0$ ,  $u, v$  are independent;  $uv = 0$  makes them a natural basis. Dimensions zero and one are immediate.

Thus  $D$  without a natural basis and with  $N(D) = \{0\}$  must have dimension at least three. Since  $D \neq E$ , the ambient dimension is at least four. The existing witness attains that bound. Consequently Barei's and Hu-Wen's sufficient hypotheses coincide in ambient dimensions at most three and first separate in dimension four. This is a proof reconstructed for the audit follow-up; no historical novelty claim is made.

## 12. Reproduce and interpret the evidence

From the package directory, run with Python 3.8+ and the standard library, without `-O`:

**python3 verify\_minimality.py --output my\_results.json**

**python3 input\_record/verify\_audit.py --output my\_original\_results.json**

The package contains both exact programs, actual results and stdout logs, a dated run record, source locations, this report and the addendum text. SHA-256 inventories identify the packaged bytes; successful replay and hash checks remain separate from mathematical correctness. The original audit is included unchanged. Nothing was published or sent to anyone during this follow-up.

Credit and scope: Andres Barei's manuscript credits Muse Spark and names Nicolás Jaramillo Torres as reviewer; the comparison acknowledges Hu and Wen. This follow-up adds a minimality proof to the earlier audit witness. No blind replication, independent human expert review, formal proof verification, historical novelty, endorsement, or causal benefit from Protocol A is claimed.